

Early Diabetes Prediction

Using Machine Learning — Pima Indians Diabetes Database

Team SK-AI | Week 4 Final Report

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Project Overview

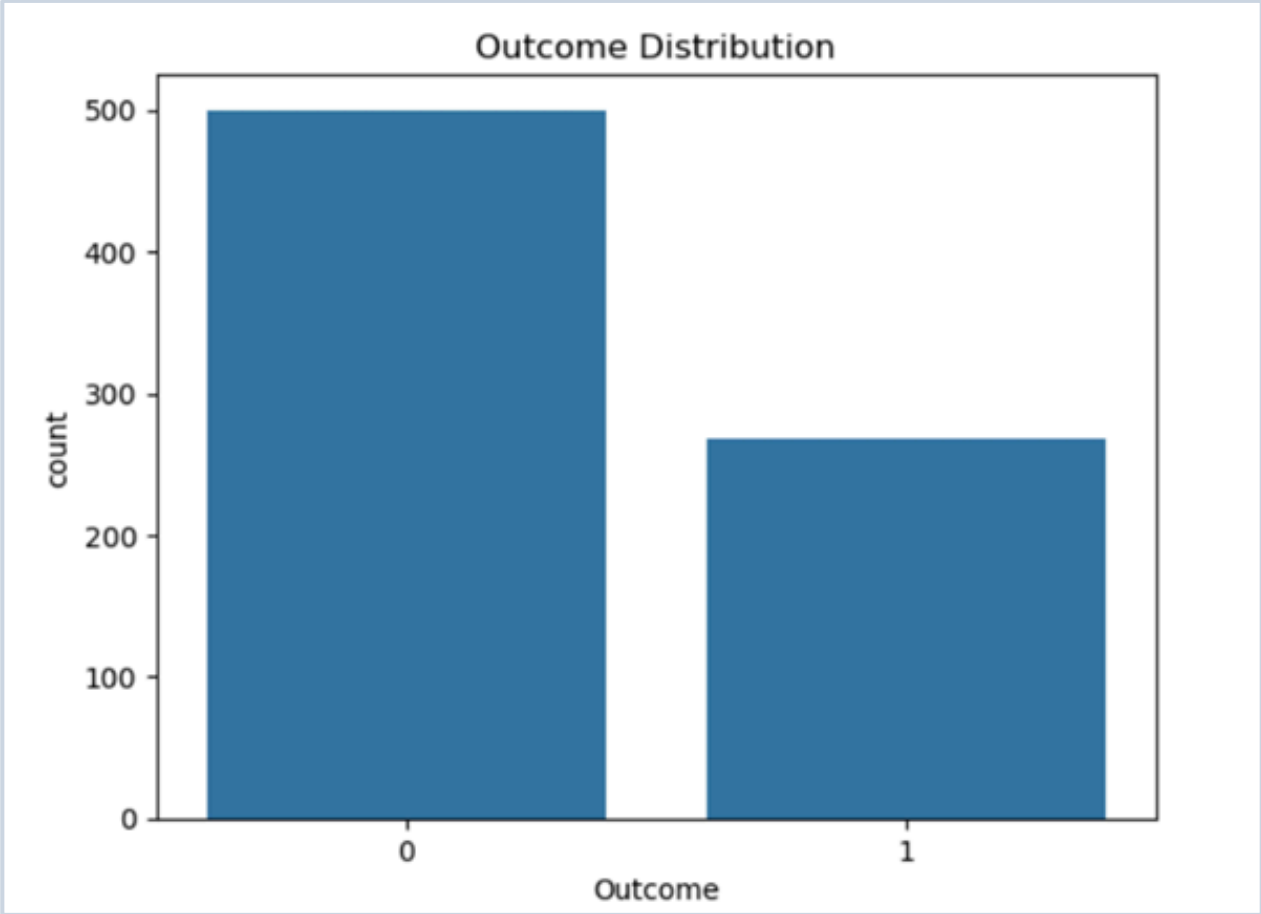
Dataset

Source:	Pima Indians Diabetes Database (Kaggle)
Records:	768 patients
Features:	8 clinical inputs + 1 target (Outcome)
Type:	Binary Classification
Target:	Outcome: 0 = No Diabetes, 1 = Diabetes

Hypothesis

Glucose and BMI will be the most significant predictors of a positive diabetes diagnosis.

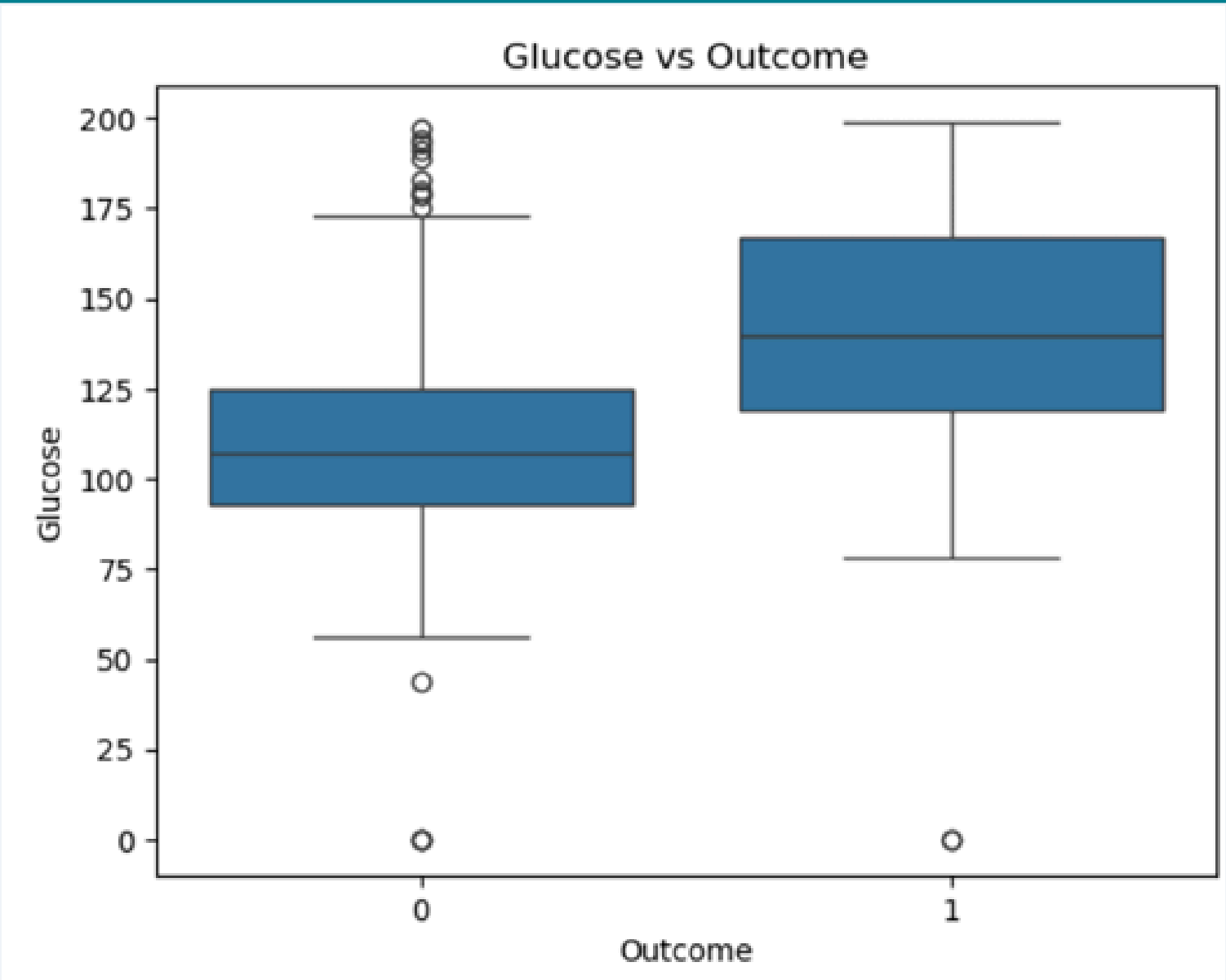
Class Distribution



⚠ Class Imbalance Present

EDA Highlights

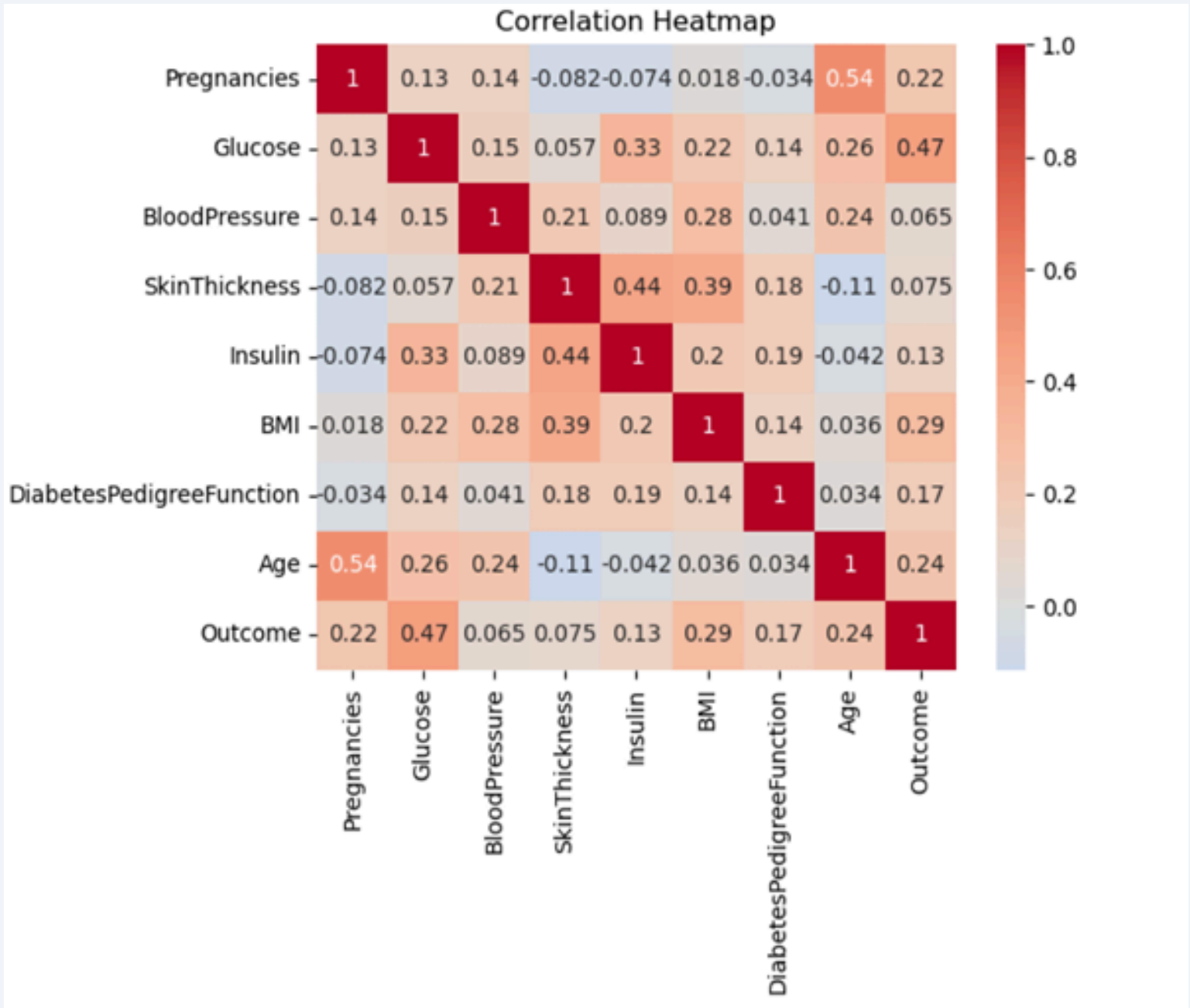
Glucose: Key Predictor



Diabetic patients average 141 mg/dL vs 110 mg/dL — a ~30-point gap confirming Glucose as the #1 predictor.

Correlation with Outcome

Glucose ($r \approx 0.47$) and BMI ($r \approx 0.29$) confirm the Week 1 hypothesis. BloodPressure & SkinThickness contribute least.



Methodology

Step 1: Preprocessing

- StandardScaler — normalize all features
- Zero values flagged (Glucose, BMI, Insulin)
- 80/20 train-test split (614 train / 154 test)
- random_state = 42 for reproducibility

Step 2: Model Selection

- Logistic Regression — linear baseline
- Decision Tree — captures non-linear rules
- K-Nearest Neighbors — distance-based
- 3 diverse algorithm types compared

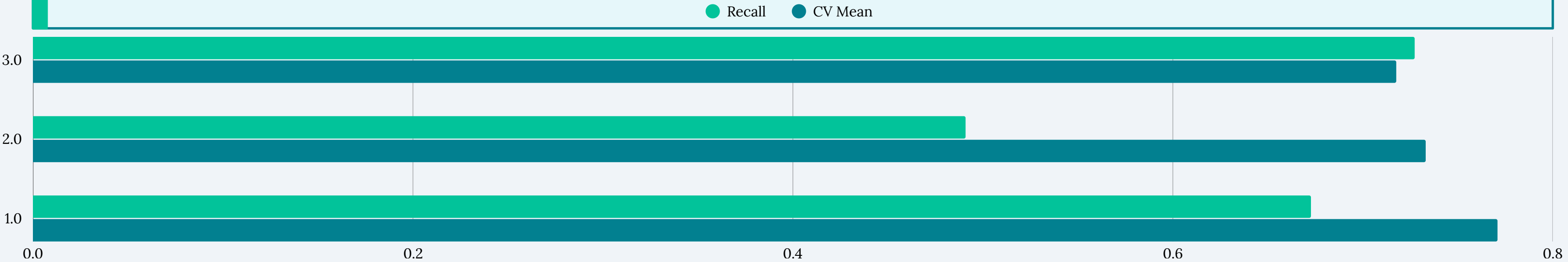
Step 3: Evaluation

- Accuracy, Precision, Recall, F1-score
- Focus on Recall (medical context)
- 5-fold Cross-Validation (CV Mean & Std)
- False Negatives = most critical error

Model Results

Model	Accuracy	Precision	Recall	F1-Score	CV Mean	CV Std
Logistic Regression	0.7532	0.6491	0.6727	0.6607	0.7709	0.0247
Decision Tree	0.7468	0.6250	0.7273	0.6723	0.7176	0.0560
KNN	0.6883	0.5745	0.4909	0.5294	0.7331	0.0201

Logistic Regression wins on stability (CV Mean 0.77, CV Std 0.025) – Decision Tree is competitive but overfits (CV Std 0.056). KNN has the lowest Recall (0.49), missing ~half of all diabetic patients.



Error Analysis

⚠ False Negatives: Logistic Regression missed ~18 out of 55 diabetic patients (33%). In a medical context, this is the most dangerous error — a sick patient goes undiagnosed.

Profile of Missed Patients

- Avg Glucose:** ~115 mg/dL (typical diabetic: ~141)
- Glucose zone:** 100–130 mg/dL — borderline range
- Other issue:** Zero values in Insulin & SkinThickness
- Root cause:** Linear boundary struggles mid-range

Where the Model Excels

- Glucose > 150 → near-certain detection
- Overall Recall = 0.67 (67% of diabetics found)
- Stable performance across 5-fold CV
- Highest overall F1-Score among all models

--- Statistics for Missed Patients (False Negatives) ---			
	Glucose	BMI	Age
count	18.000000	18.000000	18.000000
mean	-0.220995	-0.265588	-0.228489
std	0.615102	1.100402	0.599692
min	-1.279882	-4.060474	-0.956462
25%	-0.575696	-0.465486	-0.722470
50%	-0.247076	-0.259242	-0.360847
75%	0.120665	0.153245	0.277311
max	0.848324	1.714352	0.915469
--- Statistics for All Diabetic Patients in Test Set ---			
	Glucose	BMI	Age
count	55.000000	55.000000	55.000000
mean	0.682164	0.354701	0.237861
std	0.907035	1.025713	0.930213
min	-1.279882	-4.060474	-1.041549
25%	-0.027996	-0.202129	-0.445935
50%	0.785730	0.229397	0.234767
75%	1.333430	0.768803	0.745293
max	2.381884	2.691630	2.872487

Reflection & Limitations

Hypothesis partially confirmed: Glucose ($r \approx 0.47$) was the dominant predictor — BMI also significant, but Age emerged as an unexpected additional factor.

⚠️ Zero Values

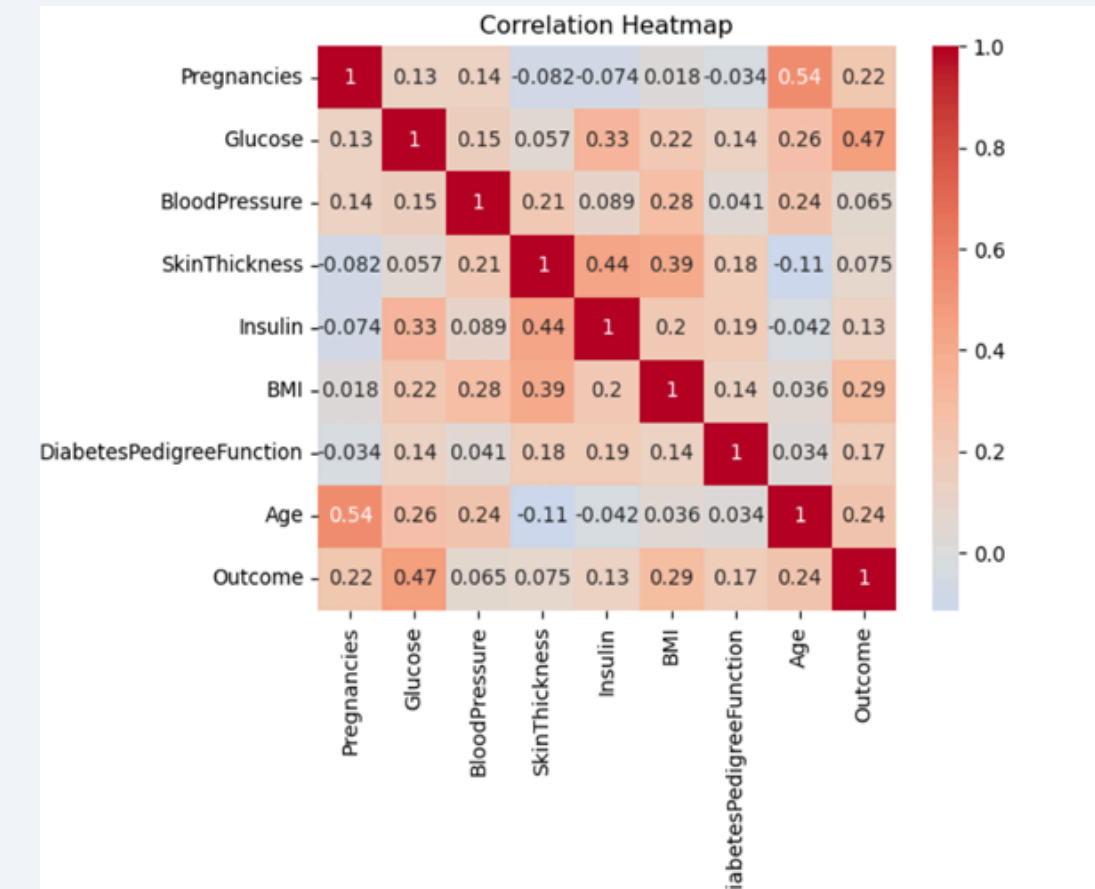
BMI, Glucose & Insulin had impossible zeros treated as valid data — degraded model performance.

⚠️ Class Imbalance

65/35 split; no SMOTE or `class_weight='balanced'` applied — biased toward majority class.

```
bio_cols = ['Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI']  
(df[bio_cols] == 0).sum()
```

```
Glucose      5  
BloodPressure 35  
SkinThickness 227  
Insulin      374  
BMI          11  
dtype: int64
```



What We'd Do Differently

Impute zeros: Replace impossible zeros with class-specific medians (diabetic vs non-diabetic)

Balance classes: Apply `class_weight='balanced'` in Logistic Regression to boost Recall on diabetic class

```
for col in ['Glucose', 'BMI', 'BloodPressure']:  
    df[col] = df[col].fillna(df[col].median())  
  
for col in ['SkinThickness', 'Insulin']:  
    df[col] = df[col].fillna(df.groupby('Outcome')[col].transform('median'))
```


Conclusion

Model	Accuracy	Precision	Recall	F1-Score	CV Mean	CV Std
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75.3%

Best Accuracy
(Logistic Reg)

0.77

CV Mean
(most stable)

33%

False Negative
Rate

0.47

Glucose–Outcome
Correlation

Logistic Regression is the best model — highest accuracy, CV stability, and F1-score

Glucose is the single most important predictor ($r \approx 0.47$) — hypothesis confirmed

Class imbalance & zero-value imputation are the two highest-priority improvements

Future work: SMOTE balancing, max_depth tuning, and formal feature importance analysis